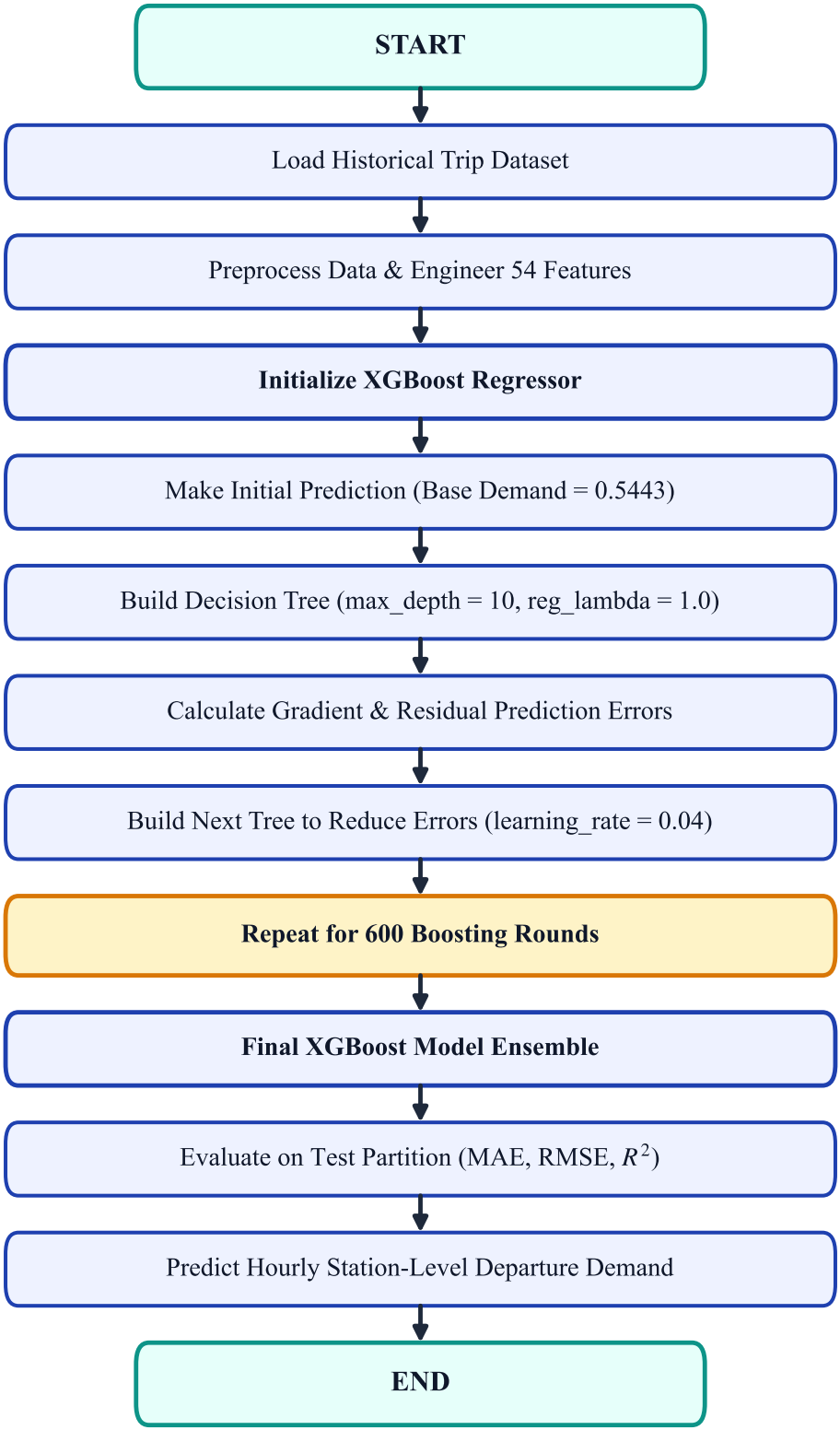
START
Load Historical Trip Dataset
Preprocess Data & Engineer 54 Features
Initialize XGBoost Regressor
Make Initial Prediction (Base Demand = 0.5443)
Build Decision Tree (max_depth = 10, reg_lambda = 1.0)
Calculate Gradient & Residual Prediction Errors
Build Next Tree to Reduce Errors (learning_rate = 0.04)
Repeat for 600 Boosting Rounds
Final XGBoost Model Ensemble
Evaluate on Test Partition (MAE, RMSE, R^2)
Predict Hourly Station-Level Departure Demand
END